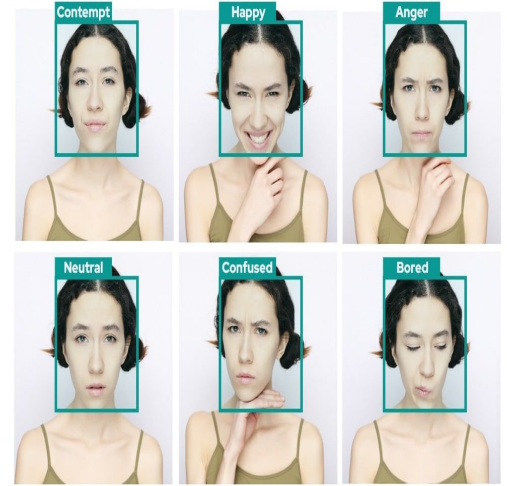


Emotion Detector

Sansshita Baskaran, Garrett Delaney, Evan Sage

Problem Statement

- Emotion recognition models are powerful but often rely on remote cloud servers to process video data
- Leads to latency, high bandwidth use, and privacy concerns
- Cloud dependence makes systems vulnerable to connectivity issues and limits real-world deployment



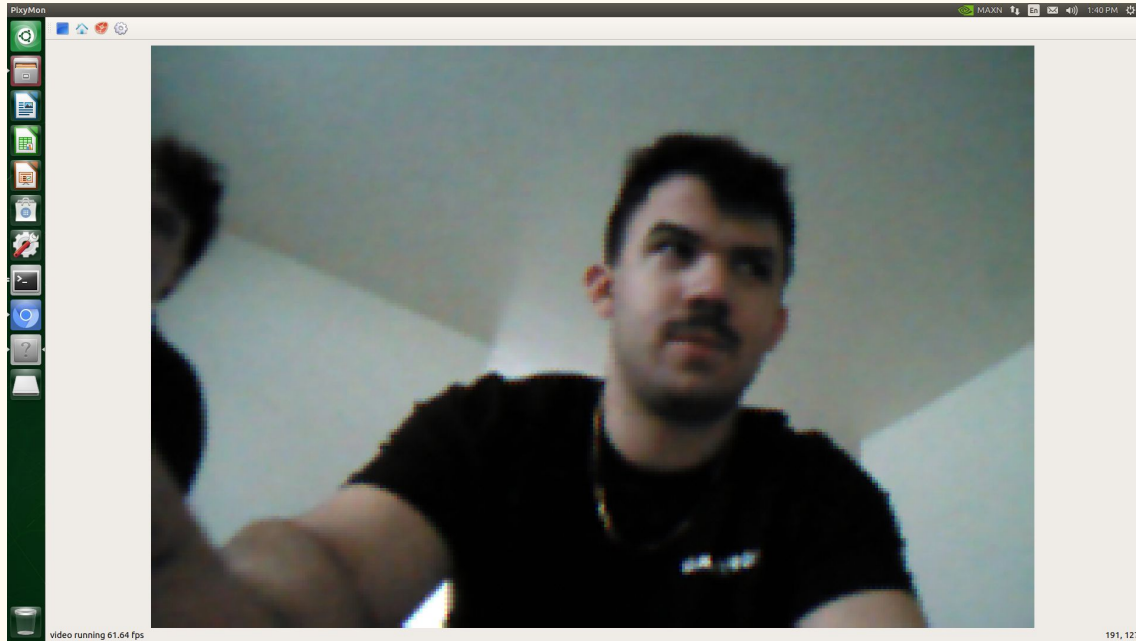
Goal and Platform

- Develop a real-time emotion detection system that works entirely on device. No internet required.
- Ensure it is fast, private, and also reliable in real-world environment.
- Platform: We chose the NVIDIA Jetson (Edge AI)

Progress

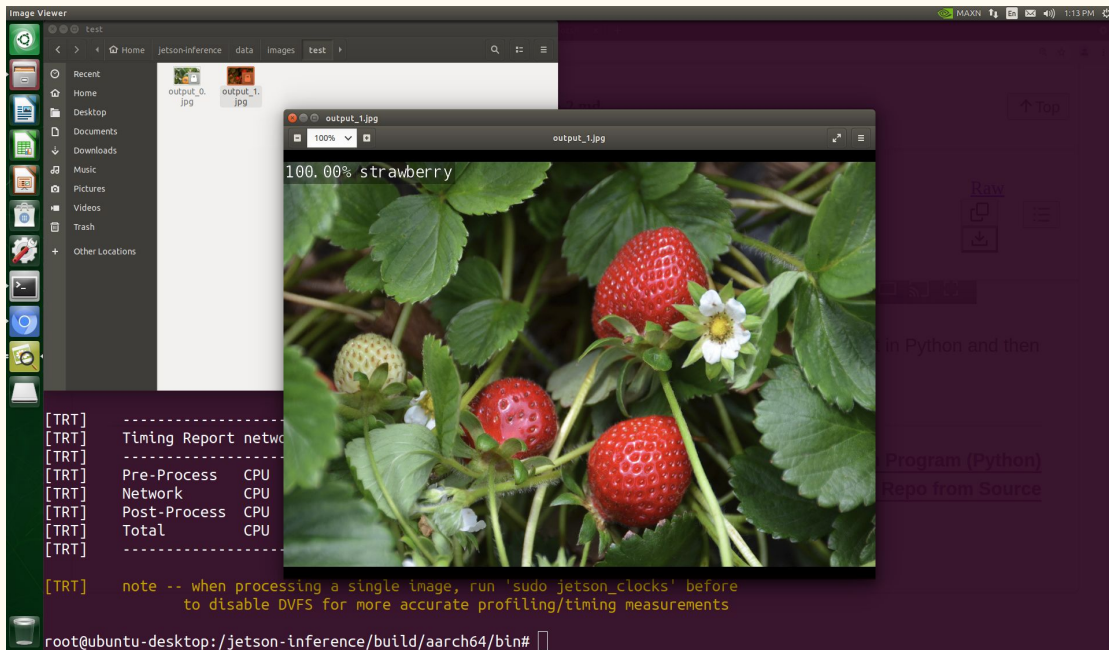
- Jetson setup complete
 - Solved Power Issues
 - Installed NVIDIA JetPack SDK
 - Verified system stability and GPU acceleration
- AI Inference Functional
 - Installed and ran Pytorch, ImageNet successfully
 - The jetson can now classify test images
- Webcam integration working
 - Connected and configured USB webcam for live video streaming.

Progress Pictures



- Webcam is connected and streaming live video directly to the Jetson.
- Demonstrates readiness for real-time emotion detection

Progress Pictures Continued



- Jetson was able to caption an image of strawberries with 100% confidence.
- Confirms Jetson-inference and GPU acceleration work locally.

Future Work

- Train the emotion recognition model using the FER-2013 dataset.
- Deploy it on the Jetson and run real-time tests with webcam
- Optimize performance with TensorRT
- Display detected emotions on-screen feedback.
- Work on Final Report and Presentation.